

Question 1:

Consider two particles moving with velocities described by vectors $\vec{v}_1 = (3; 1)$ m/s and $\vec{v}_2 = (1; 3)$ m/s. At time $t = 0$ s, they were located at points $\vec{r}_{10} = (-2; 0)$ m and $\vec{r}_{20} = (0; -3)$ m. Determine the vector of their relative position as a function of time, and calculate time and distance when they are closest to each other.

Solution:

$$\vec{r}_{1t} = (-2 + 3t; t) \text{ m}$$

$$\vec{r}_{2t} = (t; -3 + 3t) \text{ m}$$

$$\vec{R}_t = \vec{r}_{2t} - \vec{r}_{1t} = \cancel{(-2 + 3t; t)} = (+2 - 2t; 2t - 3)$$

$$|\vec{R}_t|^2 = \cancel{(-2 + 3t)^2 + (2t - 3)^2} = \cancel{4 - 12t + 9t^2 + 4t^2 - 12t + 9}$$

$$\cancel{= 13 - 24t + 13t^2}$$

$$|\vec{R}_t|^2 = (2 - 2t)^2 + (2t - 3)^2 = 4 - 8t + 4t^2 + 4t^2 - 12t + 9 \\ = 8t^2 - 20t + 13$$

$$\frac{d}{dt} |\vec{R}_t|^2 = 16t - 20 = 0 \Rightarrow t_{\text{closest}} = 5/4$$

$$|\vec{R}_t| = \sqrt{\cancel{8} \frac{25}{2} - \cancel{20} \frac{5}{4} + 13} = \sqrt{\frac{25}{2} - 25 + 13} = \sqrt{13 - \frac{25}{2}} = \sqrt{\frac{26 - 25}{2}} = \sqrt{\frac{1}{2}}$$

Question 2:

You are throwing a ball into the wind, and the wind is blowing in the same direction as the throw of the ball. The wind exerts a constant horizontal acceleration on the ball, represented as $a = \frac{g}{\sqrt{3}}$, where g is the gravitational acceleration. This acceleration acts in the direction of the ball's motion along the x-axis. You initially launch the ball with a velocity of v_0 . Your goal is to find the launch angle α that will allow the ball to travel the farthest horizontal distance possible.

Solution:

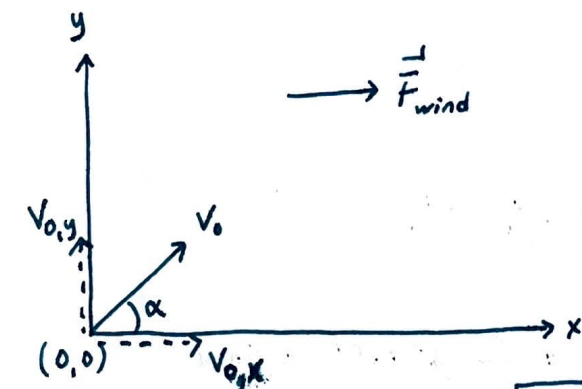
In y-direction:

$$-mg = ma$$

$$\frac{d}{dt} \dot{y} = -g \Rightarrow v(t) = -gt + v_{0,y}$$

When $v(t) = 0$, it gives us half of the total flight time:

$$t_{\text{flight}} = \frac{2 v_{0,y}}{g} \rightarrow \text{It will not be affected by } \vec{F}_{\text{wind}}.$$



In x-direction:

$$\vec{F}_{\text{wind}} = m \ddot{x} = m \frac{g}{\sqrt{3}}$$

$$\dot{x} = \frac{gt}{\sqrt{3}} + v_{0,x}$$

$$x(t) = \frac{1}{2\sqrt{3}} g t^2 + v_{0,x} t + 0$$

$$X_{\text{max}} = x\left(\frac{2 v_{0,y}}{g}\right) = \frac{1}{\sqrt{3}} \frac{v_{0,y}^2}{g} + v_{0,x} \frac{2 v_{0,y}}{g}$$

$$= \frac{2 v_{0,y}^2}{\sqrt{3} g} + \frac{2 v_{0,y} v_{0,x}}{g} = \frac{2}{\sqrt{3} g} v_0^2 \sin^2 \alpha + \frac{2 v_0^2 \sin \alpha \cos \alpha}{g} = \frac{2}{\sqrt{3} g} v_0^2 \sin^2 \alpha + \frac{v_0^2 \sin(2\alpha)}{g}$$

$$= \frac{v_0^2}{g} \left(\frac{2}{\sqrt{3}} \sin^2 \alpha + \sin(2\alpha) \right)$$

$$\frac{d}{d\alpha} [X_{\text{max}}(\alpha)] = \frac{v_0^2}{g} \left(\frac{2}{\sqrt{3}} \underbrace{2 \sin \alpha \cos \alpha}_{\sin(2\alpha)} + 2 \cos(2\alpha) \right) = 0$$

$$\Rightarrow \frac{2}{\sqrt{3}} \sin(2\alpha) + 2 \cos(2\alpha) = 0 \Rightarrow \frac{\sin(2\alpha)}{\sqrt{3}} = -\cos(2\alpha) \Rightarrow \tan(2\alpha) = -\sqrt{3}$$

$$\arctan(-\sqrt{3}) = -\pi/3$$

or $2\alpha = 120^\circ$

$$\alpha = 30^\circ$$

Question 3:

A projectile with a mass $m = 10 \text{ g}$ strikes a wall with a velocity $\vec{v}_1 = 60 \text{ m/s}$ and exits from the other side with a velocity $\vec{v}_2 = 40 \text{ m/s}$ after penetrating the wall of thickness $d = 20 \text{ cm}$. What was the average force F acting on the projectile inside the wall? Use your knowledge of dynamics!

Solution:

$$T_1 - T_2 = \frac{1}{2} m (v_1^2 - v_2^2) \rightarrow \text{loss of K.E} = \text{work done by the wall.}$$

$$\int_0^d F_{\text{avg}} dx = \frac{1}{2} m (v_1^2 - v_2^2)$$

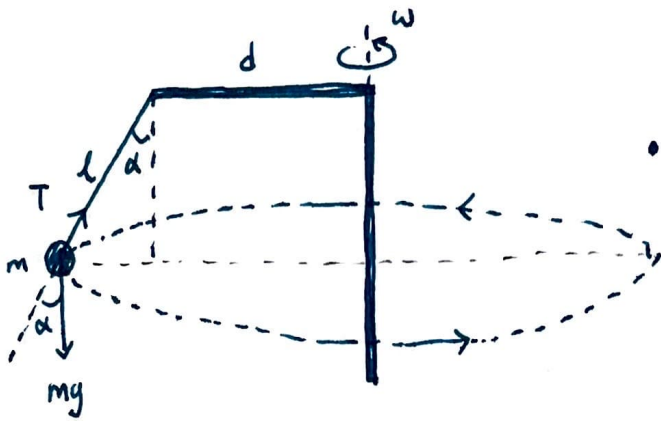
$$F_{\text{avg}} d = \frac{1}{2} m (v_1^2 - v_2^2) \Rightarrow F_{\text{avg}} = \frac{m}{2d} (v_1^2 - v_2^2)$$

$$F_{\text{avg}} = \frac{10 \times 10^{-3}}{2 \times 20 \times 10^{-2}} (60^2 - 40^2) \frac{\text{kg}}{\text{m}} \frac{\text{m}^2}{\text{s}^2} = \frac{1}{40} 20 \times 100 = 50 \text{ N}$$

Question 4:

A mathematical pendulum with a length of l is attached to the arm of a rotating carousel at a distance d from its center. The pendulum is deflected from its vertical position by an angle α . Calculate the period of rotation for the carousel.

Solution:



- The mass does a circular motion with radius:

$$r = l \sin \alpha + d$$

- The sin component of T is responsible from the circular motion (centripetal force)

$$F = ma \text{ for circular motion}$$

$$T \sin \alpha = m \frac{v^2}{r} = m \omega^2 r \quad v = \omega r$$

$$\Rightarrow T \sin \alpha = m \omega^2 (l \sin \alpha + d)$$

- Now all we need to do is to write T in terms of the other parameters:

$$T = mg \cos \alpha$$

$$\Rightarrow mg \cos \alpha \sin \alpha = m \omega^2 (l \sin \alpha + d)$$

$$\omega = \sqrt{\frac{g \sin \alpha \cos \alpha}{l \sin \alpha + d}} = 2\pi f \Rightarrow f = \frac{1}{2\pi} \sqrt{\frac{g \sin \alpha \cos \alpha}{l \sin \alpha + d}}$$

↑ rotation frequency

$$T = \frac{1}{f} = 2\pi \sqrt{\frac{l \sin \alpha + d}{g \sin \alpha \cos \alpha}} = 2\pi \sqrt{\frac{2(l \sin \alpha + d)}{g \sin(2\alpha)}}$$